

DEFINITE INTEGRATION

- If $\int f(x)dx = F(x) + C$, then $\int_a^b f(x)dx = F(b) - F(a)$
- $\int_a^a f(x)dx = 0$
- $\int_a^b f(x)dx = \int_a^b f(y)dy = \int_a^b f(z)dz = \int_a^b f(p)dp = \dots$
- $\int_a^b f(x)dx = -\int_b^a f(x)dx$
- $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$ where $a < c < b$
- $\int_a^b f(x)dx = \int_a^{c_1} f(x)dx + \int_{c_1}^{c_2} f(x)dx + \int_{c_2}^{c_3} f(x)dx + \dots + \int_{c_n}^b f(x)dx$ where $a < c_1 < c_2 < \dots < c_n < b$.
- $\int_0^{2a} f(x)dx = \int_0^a (f(x) + f(2a-x))dx$.
- $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$
- $\int_0^a f(x)dx = \int_0^a f(a-x)dx$
- $\int_a^b f(x)dx = 0$ if $f(a+x) = -f(b-x)$
- $\int_a^b f(x)dx = 2 \int_a^{\frac{a+b}{2}} f(x)dx$ if $f(a+x) = f(b-x)$
- $\int_0^a f(x)dx = \int_0^{a/2} f(x)dx + \int_0^{a/2} f(a-x)dx$
- $\int_{-a}^a f(x)dx = \begin{cases} 2 \int_0^a f(x)dx & f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0 & f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$
- $\int_0^{2a} f(x)dx = \begin{cases} 2 \int_0^a f(x)dx & f(2a-x) = f(x) \\ 0 & f(2a-x) = -f(x) \end{cases}$

- $\int_a^b xf(x)dx = \frac{a+b}{2} \int_a^b f(x)dx$ if $f(a+b-x) = f(x)$
- $\int_0^a xf(x)dx = \frac{a}{2} \int_0^a f(x)dx$ if $f(a-x) = f(x)$
- $\int_a^b f(x)dx = (b-a) \int_0^1 f((b-a)x+a)dx$
- If $f(x)$ is a periodic function with period T , then $\int_0^{nT} f(x)dx = n \int_0^T f(x)dx$ for $n \in N$
- If $f(x)$ is a periodic function with period T , then $\int_a^{a+nT} f(x)dx$ for $n \in N$ is independent of a .
- If $f: R \rightarrow R$ is a continuous periodic function with period T and $a \in R$ and $n \in N$ then

$$\int_a^{a+nT} f(x)dx = n \int_a^{a+T} f(x)dx = n \int_0^T f(x)dx$$
- If $f(x)$ is a periodic function with period T , $a \in R^+$, then

$$\int_{a+T}^{b+T} f(x)dx = \int_a^b f(x)dx$$

$$\int_{a+nT}^{b+nT} f(x)dx = \int_a^b f(x)dx \text{ where } n \in N$$

$$\int_{mT}^{nT} f(x)dx = (n-m) \int_0^T f(x)dx \text{ where } m, n \in N$$
- **Leibnitz Rule:**
 - (1) If $f(x)$ is continuous on $[a, b]$ and $\alpha(x)$ and $\beta(x)$ are differentiable functions of x whose values lies on $[a, b]$, then

$$\frac{d}{dx} \left(\int_{\beta(x)}^{\alpha(x)} f(t)dt \right) = f(\alpha(x))\alpha'(x) - f(\beta(x))\beta'(x).$$
 - (2) If $\alpha(x)$ and $\beta(x)$ are defined on $[a, b]$ and differentiable at a point $x \in [a, b]$ and $f(x, t)$ is continuous, then:

$$\frac{d}{dx} \left(\int_{\beta(x)}^{\alpha(x)} f(x, t)dt \right) = \int_{\beta(x)}^{\alpha(x)} \frac{\partial}{\partial x} f(x, t)dt + \frac{d\alpha(x)}{dx} f(x, \alpha(x)) - \frac{d\beta(x)}{dx} f(x, \beta(x)).$$
- If $f(x) \geq 0$ in $[a, b]$ then $\int_a^b f(x)dx \geq 0$.
- If a function $f(x)$ is integrable and non-negative on $[a, b]$ and there exists appoint $c \in [a, b]$ of continuity of $f(x)$ for which $f(c) > 0$, then $\int_a^b f(x)dx > 0$.

- If $f(x) \leq g(x)$ in $[a, b]$ then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.
- $\left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx$.
- If m, M are smallest and greatest values of a function $f(x)$ then $m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$.
- **Schwarz – Bunyakovsky inequality:** For any two functions $f(x)$ and $g(x)$, integrable on the interval $[a, b]$, the Schwarz-Bunyakovsky inequality hold i.e. $\left| \int_a^b f(x)g(x)dx \right| \leq \sqrt{\int_a^b f^2(x)dx} \times \sqrt{\int_a^b g^2(x)dx}$.
- If a function $f(x)$ is continuous on the interval $[a, b]$, then there exist a point $c \in [a, b]$ such that $\int_a^b f(x)dx = f(c)(b-a)$, $a < c < b$.
- **Mean Value or Average value of a function:** Mean value or average value of a function $f(x)$ on the interval $[a, b]$, is given by $f(x) \geq \frac{1}{b-a} \int_a^b f(x)dx$.
- **Root Mean Square Value (RMSV):** Root Mean square value of a function $f(x)$ on the interval (a, b) is given by $\sqrt{\frac{\int_a^b (f(x))^2 dx}{b-a}}$.
- **Holder's Inequality for Integrals:** $\left| \int_a^b f(x)g(x)dx \right| \leq \left\{ \int_a^b |f(x)|^p dx \right\}^{1/p} \times \left\{ \int_a^b |g(x)|^q dx \right\}^{1/q}$
where $\frac{1}{p} + \frac{1}{q} = 1$, $p > 1, q > 1$. If $p = q = 2$, Holder's Inequality reduces to Cauchy-Schwarz Inequality for integral. The equality hold if and only if $\frac{|f(x)|^{p-1}}{|g(x)|}$ is a constant.
- **Minkowski's Inequality for Integrals:** If $p > 1$, $\left\{ \int_a^b |f(x) + g(x)|^p dx \right\}^{1/p} \leq \left\{ \int_a^b |f(x)|^p dx \right\}^{1/p} + \left\{ \int_a^b |g(x)|^p dx \right\}^{1/p}$. The equality hold if and only if $\frac{f(x)}{g(x)}$ is a constant.

- **Second Fundamental Theorem:** If $f(x)$ is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)dt$ is

differentiable at every point x in $[a, b]$ and $\frac{dF(x)}{dx} = \frac{d}{dx} \int_a^x f(t)dt = f(x)$.

- **Walli's Formula:** $\int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx = \begin{cases} \frac{(n-1)(n-3)(n-5)\dots 1}{n(n-2)(n-4)\dots 2} \times \frac{\pi}{2} & \text{if } n \text{ is even} \\ \frac{(n-1)(n-3)(n-5)\dots 2}{n(n-2)(n-4)\dots 3} & \text{if } n \text{ is odd} \end{cases}$

- $\int_0^{\pi/4} \tan^n x dx = \begin{cases} \frac{1}{n-1} - \frac{1}{n-3} + \frac{1}{n-5} - \dots + (-1)^{\frac{n+1}{2}} \frac{1}{2} + (-1)^{\frac{n-1}{2}} \frac{\pi}{4} & \text{if } n \text{ is even} \\ \frac{1}{n-1} - \frac{1}{n-3} + \frac{1}{n-5} - \dots + (-1)^{\frac{n+2}{2}} \frac{1}{2} + (-1)^{\frac{n}{2}} \frac{1}{2} \ln 2 & \text{if } n \text{ is odd} \end{cases}$

- $\int_0^{\pi/4} (\tan^n x + \tan^{n-2} x) dx = \frac{1}{n-1}$

- $\int_0^{\pi/4} (\cot^n x + \cot^{n-2} x) dx = \frac{1}{n-1}$

- $I_n = \int_0^{\pi/4} \sec^n x dx = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}$

- $\int_0^{\pi/2} \sin^m x \cos^n x dx = \begin{cases} \frac{(m-1)(m-3)\dots 1 \times (n-1)(n-3)\dots 1}{(m+n)(m+n-2)(m+n-4)\dots 2} \times \frac{\pi}{2} & \text{if } m, n \text{ both are even} \\ \frac{(m-1)(m-3)\dots \times (n-1)(n-3)\dots}{(m+n)(m+n-2)(m+n-4)\dots} & \text{otherwise} \end{cases}$

- $\int_0^{\pi/2} \sin^m x \cos x dx = \left[\frac{\sin^{m+1} x}{m+1} \right]_0^{\pi/2} = \frac{1}{m+1}$.

- $\int_0^{\pi/2} \cos^m x \sin x dx = - \left[\frac{\cos^{m+1} x}{m+1} \right]_0^{\pi/2} = \frac{1}{m+1}$.

- $\int_0^{\pi} \sin^m x \cos^n x dx = 0$, when n the power of $\cos x$ is odd.

- $\int_0^{\pi} \sin^n x dx = 2 \int_0^{\pi/2} \sin^n x dx$ for all positive integral value of n .

- $\int_a^b \sqrt{\frac{x-a}{b-x}} dx = \frac{\pi}{2} (b-a)$

- $\int_a^b \sqrt{(x-a)(b-x)} dx = \frac{\pi}{8} (b-a)^2$

- $\int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx = \textcircled{R}$
- $\int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}$
- $\int_b^a \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{a+b}{2}$
- $\int_0^{\pi/2} \frac{f(\sin x)}{f(\sin x) + f(\cos x)} dx = \frac{\textcircled{R}}{4}$
- $\int_0^{\pi/2} \frac{f(\tan x)}{f(\tan x) + f(\cot x)} dx = \frac{\textcircled{R}}{4}$
- $\int_0^{\pi/2} \frac{f(\sec x)}{f(\sec x) + f(\operatorname{cosec} x)} dx = \frac{\textcircled{R}}{4}$
- $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx = (a+b) \frac{\textcircled{R}}{4}$
- $\int_0^{\pi/2} \frac{a \tan x + b \cot x}{\tan x + \cot x} dx = (a+b) \frac{\textcircled{R}}{4}$
- $\int_0^{\pi/2} \frac{a \sec x + b \operatorname{cosec} x}{\sec x + \operatorname{cosec} x} dx = (a+b) \frac{\textcircled{R}}{4}$
- $\int_0^{\pi/2} \ln(\sin x) dx = \int_0^{\pi/2} \ln(\cos x) dx = -\frac{\textcircled{R}}{2} \ln 2$
- $\int_0^{\pi/2} \ln(\tan x) dx = \int_0^{\pi/2} \ln(\tan x) dx = 0$
- $\int_0^{\pi/2} \frac{1}{a \cos^2 x + b \sin^2 x} dx = \frac{\textcircled{R}}{2ab}$
- $\int_0^\infty \frac{1}{(x + \sqrt{x^2 - 1})^n} dx = \frac{n}{n^2 - 1}$
- $\int_0^n [x] dx = \frac{n(n-1)}{2}$, where $n \in N$
- $\int_{-n}^0 [x] dx = -\frac{n(n+1)}{2}$, where $n \in N$
- $\int_{-n}^n [x] dx = -n$, where $n \in N$

• **INTEGRAL AS A LIMIT OF SUM:**

$$(1) \quad \int_a^b f(x)dx = \lim_{n \rightarrow \infty} h [f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h)] \text{ where } h = \frac{b-a}{n}$$

$$(2) \quad \int_a^b f(x)dx = \lim_{n \rightarrow \infty} h [f(a+h) + f(a+2h) + \dots + f(a+nh)] \text{ where } h = \frac{b-a}{n}$$

$$(3) \quad \int_0^a f(x)dx = \lim_{n \rightarrow \infty} h [f(0) + f(h) + f(2h) + \dots + f((n-1)h)] \text{ where } h = \frac{a}{n}$$

$$(4) \quad \int_0^a f(x)dx = \lim_{n \rightarrow \infty} h [f(h) + f(2h) + \dots + f(nh)] \text{ where } h = \frac{a}{n}$$

$$(5) \quad \int_a^b f(x)dx = \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} f(a+rh) \text{ where } h = \frac{b-a}{n}$$

$$(6) \quad \int_a^b f(x)dx = \lim_{n \rightarrow \infty} h \sum_{r=1}^n f(a+rh) \text{ where } h = \frac{b-a}{n}$$

$$(7) \quad \int_0^a f(x)dx = \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} f(rh) \text{ where } h = \frac{a}{n}$$

$$(8) \quad \int_0^a f(x)dx = \lim_{n \rightarrow \infty} h \sum_{r=1}^n f(rh) \text{ where } h = \frac{a}{n}$$

$$(9) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right) = \int_0^1 f(x)dx$$

$$(10) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x)dx$$

$$(11) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{a(n-1)} f\left(\frac{r}{n}\right) = \int_0^a f(x)dx$$

$$(12) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{an} f\left(\frac{r}{n}\right) = \int_0^a f(x)dx$$

• **Improper Integral:**

$$(1) \quad \int_a^\infty f(x)dx = \lim_{b \rightarrow \infty} \int_a^b f(x)dx$$

$$(2) \quad \int_{-\infty}^b f(x)dx = \lim_{a \rightarrow -\infty} \int_a^b f(x)dx$$

$$(3) \quad \int_{-\infty}^\infty f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^\infty f(x)dx = \lim_{b \rightarrow \infty} \int_a^b f(x)dx + \lim_{b \rightarrow \infty} \int_c^b f(x)dx.$$

• **Beta and Gamma Function:**

$$(1) \quad \text{Beta Function/First Eulerian Integral: } B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad m > 0, n > 0.$$

$$(2) \quad \text{Gamma Function/Second Eulerian Integral: } \Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx, \quad n > 0.$$

• **Properties of Beta and Gamma Function:**

$$(1) \quad \Gamma(n+1) = \int_0^\infty e^{-x} x^n dx = n \int_0^\infty e^{-x} x^{n-1} dx = n\Gamma(n).$$

$$(2) \quad \text{When } n \text{ is a positive integer, then } \Gamma(n+1) = n!.$$

$$(3) \quad \int_0^\infty e^{-kx} x^{n-1} dx = \frac{\Gamma(n)}{k^n}, \quad k > 0, n > 0.$$

$$(4) \quad B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$

$$(5) \quad \Gamma(m) \times \Gamma(1-m) = \frac{\pi}{\sin m\pi}, \quad 0 < m < 1.$$

$$(6) \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

$$(7) \quad B(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx.$$

$$(8) \quad \int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}, \quad m > -1, n > -1.$$

• **Properties of Curve Tracing:** Let a curve $f(x, y) = 0$ is in Cartesian form, then

(a) **Symmetry:**

- (1) The curve $f(x, y) = 0$ is symmetric about y-axis if $f(-x, y) = f(x, y)$.
- (2) The curve $f(x, y) = 0$ is symmetric about x-axis if $f(x, -y) = f(x, y)$.
- (3) The curve $f(x, y) = 0$ is symmetric about origin if $f(-x, -y) = f(x, y)$.
- (4) The curve $f(x, y) = 0$ is symmetric about the line $y = x$ if $f(x, y) = f(y, x)$.
- (5) The curve $f(x, y) = 0$ is symmetric about the line $y = -x$ if $f(x, y) = f(-y, -x)$.

- (b) **Origin:** If there is no constant term in $f(x, y) = 0$, then the curve passes through the origin. Then the tangent at origin is given by equating the lowest degree term of the curve to zero.
- (c) **Asymptotes:** Asymptote to the curve is a line nearest to origin and tangent at a point farthest from the origin.
- (1) The vertical asymptotes or the asymptote parallel to y-axis of the given curve can be obtained by equating the coefficient of highest power of y (provided that is not a constant) to zero.
A line $x = a$ is a vertical asymptote to the function $y = f(x)$ if $\lim_{x \rightarrow a^+} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^-} f(x) = \pm\infty$.
- (2) The horizontal asymptotes or the asymptote parallel to x-axis of the given curve can be obtained by equating the coefficient of highest power of x (provided that is not a constant) to zero.
A line $y = a$ is a horizontal asymptote to the function $y = f(x)$ if $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = b$.
- (d) **Region:** Find out the regions of the plane in which no part of the curve lies. To determine such regions we solve the given equation for y in terms of x or x in terms of y. Let y becomes imaginary for $x > a$, then curve does not lie in the region $x > a$.
- (e) **Critical Points:** These are the points at which either the derivative of the function is zero or it does not exist. So find out the value of x at which $\frac{dy}{dx} = 0$. At such points generally y changes its character from an increasing function of x to a decreasing function of x or vice versa.

• **AREA BOUNDED BY REGIONS BY USING INTEGRATION:**

- (1) Area of region bounded by the curve $y = f(x)$, x-axis and the line $x = a, x = b$ is $\left| \int_a^b y dx \right| = \left| \int_a^b f(x) dx \right|$ if curve does not cut x-axis between $x = a, x = b$.
- (2) Area of region bounded by the curve $x = f(y)$, y-axis and the line $y = a, y = b$ is $\left| \int_a^b x dy \right| = \left| \int_a^b f(y) dy \right|$ if curve does not cut y-axis between $y = a, y = b$.
- (3) If $f(x) > 0, \forall x \in [a, c]$ and $f(x) < 0, \forall x \in [c, b]$, then area bounded by the curve $y = f(x)$, x-axis and the line $x = a, x = b$ is $\int_a^c y dx - \int_c^b y dx = \int_a^c f(x) dx - \int_c^b f(x) dx$.
- (4) If $f(y) > 0, \forall y \in [a, c]$ and $f(y) < 0, \forall y \in [c, b]$, then area bounded by the curve $x = f(y)$, y-axis and the line $y = a, y = b$ is $\int_a^c x dy - \int_c^b x dy = \int_a^c f(y) dy - \int_c^b f(y) dy$.

- (5) Let $y = f(x)$ and $y = g(x)$ are two curves, then area bounded by the two curves and line $x = a, x = b$ is $\left| \int_a^b (f(x) - g(x)) dx \right|$.
- (6) Let $y = f(x)$ and $y = g(x)$ are two curves intersect at $x = c$, ($a < c < b$), then area bounded by the two curves and line $x = a, x = b$ is $\left| \int_a^c (f(x) - g(x)) dx \right| + \left| \int_c^b (f(x) - g(x)) dx \right|$.
- (7) Area bounded by $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16ab}{3}$ sq. units.
- (8) Area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\pi.ab$ sq. units.
- (9) Area bounded by the circle $x^2 + y^2 = a^2$ is $\pi.a^2$ sq. units.
- (10) Area bounded by the one arch of $y = \sin ax$ or $y = \cos ax$ and x axis is $\frac{2}{a}$ sq. units.
- (11) Area bounded by the one arch of $y = \sin ax$ or $y = \cos ax$ and x axis in $[0, n\pi]$ is $\frac{2n}{a}$ sq. units.
- (12) Area bounded by the curve $y^2 = 4ax$ and $y = mx$ is $\frac{8a^2}{3m^3}$ sq. units.
- (13) Area bounded by the curve $x^2 = 4ay$ and $y = mx$ is $\frac{8a^2m^3}{3}$ sq. units.
- (14) Area bounded by the curve $y^2 = 4ax$ and its latus rectum $x = a$ is $\frac{8a^2}{3}$ sq. units.
- (15) Area bounded by the curve $y^2 = 4ax$ and line $x = b$ is $\frac{8a^{1/2}b^{3/2}}{3}$ sq. units.
- (16) Area bounded by the curve $x^2 = 4ay$ and its latus rectum $y = a$ is $\frac{8a^2}{3}$ sq. units.
- (17) Area bounded by the curve $x^2 = 4ay$ and line $y = b$ is $\frac{8a^{1/2}b^{3/2}}{3}$ sq. units.
- (18) Area bounded by the curve $y^2 = 4ax$ and line $x = b, x = c$ is $\frac{8a^{1/2}(c^{3/2} - b^{3/2})}{3}$ sq. units.
- (19) Area bounded by $y = ax^2 + bx + c$ and the line $y = mx + k$ is $\frac{\Delta^{3/2}}{6a^2}$ sq. units where Δ be the discriminant of $ax^2 + (b - m)x + (c - k)$. i.e $\Delta = \sqrt{(b - m)^2 - 4a(c - k)}$.
- (20) Area bounded by $x = ay^2 + by + c$ and the line $x = my + k$ is $\frac{\Delta^{3/2}}{6a^2}$ sq. units where Δ be the discriminant of $ay^2 + (b - m)y + (c - k)$. i.e $\Delta = \sqrt{(b - m)^2 - 4a(c - k)}$.

- **Parametric Form for the area under curve:** Let the curve given by $x = f(t), y = g(t)$. If on eliminating t , y can be expressed as a function of x , then the area can be obtained by the formula. If t cannot be eliminated then, $\text{Area} = \int_a^b y dx = \int_{t_1}^{t_2} g(t) f'(t) dt$ where $a = f(t_1), b = f(t_2)$.
- **Area in Polar Co-ordinates:** Let $r = f(\theta)$ be a curve, where $f(\theta)$ is finite, continuous and single valued function in the interval $\alpha < \theta < \beta$. The area bounded by the curve and the radii vectors $\theta = \alpha, \theta = \beta$ is given by $\text{Area} = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2} \int_{\alpha}^{\beta} (f(\theta))^2 d\theta$.
